RAJALAKSHMI ENGINEERING COLLEGE

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C2

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|---|---|
| **Started on** | Wednesday, 7 October 2026, 10:16 PM |
| **State** | Finished |
| **Completed on** | Wednesday, 7 October 2026, 10:16 PM |
| **Time taken** | 38 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

**For example:**

| Input | Result |
|---|---|
| 3<br>1 3 5<br>4 | 1 |

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int has_pair_with_diff(int A[], int n, int k) {
    int i = 0;
    int j = 1;

    // Use two pointers to find the difference
    while (i < n && j < n) {
        int diff = A[j] - A[i];

        // Ensure we are looking at two different indices
        if (diff == k && i != j) {
            return 1; // Pair found
        }
        else if (diff < k) {
            j++; // Increase the difference by moving the right pointer
        }
        else {
            i++; // Decrease the difference by moving the left pointer
            if (i == j) {
                j++; // Ensure pointers don't overlap
            }
        }
    }
    return 0; // No pair found
}

int main() {
    int n, k;

    // Read the number of elements
```

```
33      if (scanf("%d", &n) != 1) return 0;
34
35      // Allocate memory for the array dynamically
36      int *A = (int *)malloc(n * sizeof(int));
37      if (A == NULL) return 1;
38
39      // Read the array elements
40      for (int i = 0; i < n; i++) {
41          if (scanf("%d", &A[i]) != 1) {
42              free(A);
43              return 0;
44          }
45      }
46
47      // Read the target difference k
48      if (scanf("%d", &k) != 1) {
49          free(A);
50          return 0;
51      }
52
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 3<br>1 3 5<br>4 | 1 | 1 | ✔ |
| ✔ | 10<br>1 4 6 8 12 14 15 20 21 25<br>1 | 1 | 1 | ✔ |
| ✔ | 10<br>1 2 3 5 11 14 16 24 28 29<br>0 | 0 | 0 | ✔ |
| ✔ | 10<br>0 2 3 7 13 14 15 20 24 25<br>10 | 1 | 1 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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